

Submission Worksheet

Submission Data

Course: IT202-008-S2026

Assignment: IT202 PHP Multi-Dimension Problems

Student: Deep S. (dns33)

Status: Submitted | **Worksheet Progress:** 100%

Potential Grade: 10.00/10.00 (100.00%)

Received Grade: 0.00/10.00 (0.00%)

Started: 3/26/2026 8:52:02 PM

Updated: 3/26/2026 9:57:10 PM

Grading Link: <https://learn.ethereallab.app/assignment/v3/IT202-008-S2026/it202-php-multi-dimension-problems/grading/dns33>

View Link: <https://learn.ethereallab.app/assignment/v3/IT202-008-S2026/it202-php-multi-dimension-problems/view/dns33>

Instructions

- Overview Link: <https://youtu.be/id5giyXReal>
- 1. Ensure you read all instructions and objectives before starting.
- 2. Create a new branch from `qa` called `M6-Homework`
 1. `git checkout qa` (ensure proper starting branch)
 2. `git pull origin qa` (ensure history is up to date)
 3. `git checkout -b M6-Homework` (create and switch to branch)
- 3. Copy the template code from here: [GitHub Repository - M6 Homework](#)
 - It includes Problems 1-3 and `base.php`. Put all into an `M6` folder or similar inside your `public_html`
 - Immediately record to history
 - ☐ `git add public_html`
 - ☐ `git commit -m "adding M6 HW baseline files"`
 - ☐ `git push origin M6-Homework`
 - ☐ Create a Pull Request from `M6-Homework` to `dev` and keep it open
- 4. Fill out the below worksheet
 - Each Problem requires the following as you work
 - ☐ Ensure there's a comment with your UCID, date, and brief summary of how the problem was solved
 - ☐ Initial outline/plan of how you'll solve it via comments (add/commit after this stage)
 - ☐ Code solution (add/commit periodically as needed)
- 5. Once finished, click "Submit and Export"
- 6. Locally add the generated PDF to a folder of your choosing inside your repository folder and move it to Github
 1. `git add .`
 2. `git commit -m "adding PDF"`
 3. `git push origin M6-Homework`
 4. On Github merge the pull request from `M6-Homework` to `qa`
 5. On Github create a pull request from `qa` to `prod` and immediately merge. (This will trigger the `prod` deploy to make the render.com prod links work)

7. Upload the same PDF to Canvas

8. Sync Local

1. git checkout qa
2. git pull origin qa

Section #1: (2.5 pts.) Problem 1 - Subset

Progress: 100%

≡ Task #1 (2.50 pts.) - Edit the `processBirds` function to extract properties

Progress: 100%

Details:

- Only make edits where noted via provided comments
- Challenge: Extract the name, color, region of each bird into the \$subset array
- Step 1: sketch out plan using comments (include ucid and date)
- Step 2: Add/commit your outline of comments (required for full credit)
- Step 3: Add code to solve the problem (add/commit as needed)

Part 1:

Progress: 100%

Details:

Two screenshots are expected

1. Snippet of relevant code showing solution (with ucid/date comment)
2. Full output of executing the program (visit the proper file on Render.com qa after a manual deploy)

```
10 // Start edits
11 // UCID: dms33
12 // Date: 2026-03-26
13 // Plan:
14 // 1) Loop through each bird in $birds.
15 // 2) For each bird, create a smaller array with only name, color, region.
16 // 3) Push that smaller array into $subset.
17 // 4) Print $subset at the end (already handled below).
18
19 for (var array{name: mixed, color: mixed, region: mixed} in $subset) {
20   $subset[] = [
21     "name" => $bird["name"],
22     "color" => $bird["color"],
23     "region" => $bird["region"]
24   ];
25 }
26
27 // End edits
28 echo "<pre>" . var_export($subset, true) . "</pre>";
```

Code snippet



qa output

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Part 2:

Progress: 100%

Details:

- Direct link to the file in the homework related branch from Github (should end in `.php` , , zero credit if it does not)
- Direct link to the file on Render.com Prod (Just grab the base prod url and manually enter the path to the file)

URL #1

https://github.com/shahdp2/IT202-SP2026/blob/M6-Homework/public_html/M6/problem1.php



URL

https://github.com/shahdp2/IT202-SP2026/blob/M6-Homework/public_html/M6/problem1.php



URL #2

<https://dns33-it202-008-prod.onrender.com/M6/problem1.php>



URL

<https://dns33-it202-008-prod.onrender.com/M6/problem1.php>



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Part 3:

Progress: 100%

Details:

Briefly explain how your solution works. Describe the logic, steps, or decisions your code makes to reach the result. Do not restate the problem or list requirements.

Your Response:

In processBirds(), I loop through the \$birds array one bird at a time. For each bird, I build a new smaller array that only keeps the name, color, and region values, then append that to \$subset. After the loop finishes, I print \$subset to verify the extracted data for each bird.

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Section #2: (2.5 pts.) Problem 2 - Adding Properties

Progress: 100%

Task #1 (2.50 pts.) - Edit the processCars function to add properties

Progress: 100%

Details:

- Only make edits where noted via provided comments
- Challenge 1: Add a new property called age that's set from today's year and the car's year
- Challenge 2: Add a new property called `isClassic` that's true/false based on `$classic_age`
- Step 1: sketch out plan using comments (include ucid and date)
- Step 2: Add/commit your outline of comments (required for full credit)
- Step 3: Add code to solve the problem (add/commit as needed)

Part 1:

Progress: 100%

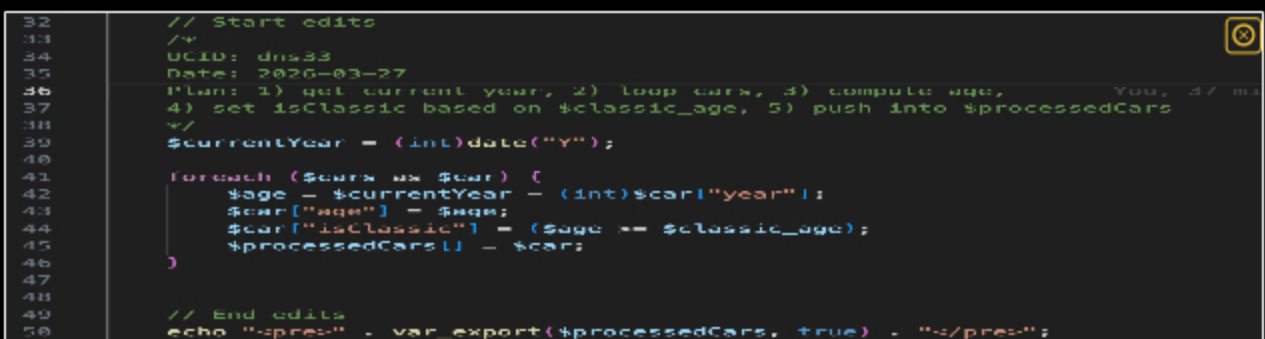
Details:

Two screenshots are expected

1. Snippet of relevant code showing solution (with ucid/date comment)
2. Full output of executing the program (visit the proper file on Render.com qa after a manual deploy)



qa output



code snippet

Part 2:

Progress: 100%

Details:

- Direct link to the file in the homework related branch from Github (should end in `.php` , , zero credit if it does not)
- Direct link to the file on Render.com Prod (Just grab the base prod url and manually enter the path to the file)

URL #1

https://github.com/shahdp2/IT202-SP2026/blob/M6-Homework/public_html/M6/problem2.php



URL

https://github.com/shahdp2/IT202-SP2026/blob/M6-Homework/public_html/M6/problem2.php



URL #2

<https://dns33-it202-008-prod.onrender.com/M6/problem2.php>



URL

<https://dns33-it202-008-prod.onrender.com/M6/problem2.php>



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Part 3:

Progress: 100%

Details:

Briefly explain how your solution works. Describe the logic, steps, or decisions your code makes to reach the result. Do not restate the problem or list requirements.

Your Response:

I first get the current year using PHP (so the math stays correct no matter when it's run). Then I loop through each car in \$cars, copy the original car data into a new array entry, and calculate age as `currentYear - carYear`. After that, I set `isClassic` to true if the computed age is greater than or equal to `$classic_age (25)`, otherwise false. Each updated car gets pushed into `$processedCars`, which is what gets printed at the end.



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Section #3: (2.5 pts.) Problem 3 - Join

Progress: 100%

≡ Task #1 (2.50 pts.) - Edit the `joinArrays` function to combine two arrays based on a common key

Progress: 100%

Details:

- Only make edits where noted via provided comments
- Challenge: Combine the data in both arrays by the `userId` property
- Step 1: sketch out plan using comments (include ucid and date)
- Step 2: Add/commit your outline of comments (required for full credit)
- Step 3: Add code to solve the problem (add/commit as needed)

Part 1:

Progress: 100%

Details:

Two screenshots are expected

1. Snippet of relevant code showing solution (with ucid/date comment)
2. Full output of executing the program (visit the proper file on Render.com qa after a manual deploy)

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52 // Start edits
53 // UCID: d0a644
54 // Date: 2026-03-27
55 // You, 25 minutes ago • Problem 3 solution
56 1) Build a lookup map from activities using userId as the key.
57 2) Loop through users; for each user, find matching activity in the map.
58 3) Merge the user fields + activity field into one array and push to joined.
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code snippet

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qa output

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Part 2:

Progress: 100%

Details:

- Direct link to the file in the homework related branch from Github (should end in `.php` , , zero credit if it does not)
- Direct link to the file on Render.com Prod (Just grab the base prod url and manually enter the path to the file)

the path to the file)

URL #1

https://github.com/shahdp2/IT202-SP2026/blob/M6-Homework/public_html/M6/problem3.php



URL

https://github.com/shahdp2/IT202-SP2026/blob/M6-Homework/public_html/M6/problem3.php



URL #2

<https://dns33-it202-008-prod.onrender.com/M6/problem3.php>



URL

<https://dns33-it202-008-prod.onrender.com/M6/problem3.php>



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Part 3:

Progress: 100%

Details:

Briefly explain how your solution works. Describe the logic, steps, or decisions your code makes to reach the result. Do not restate the problem or list requirements.

Your Response:

I create a lookup array (\$activityMap) where each activity is stored by its userId so I can find matches instantly. Then I loop through the users, grab the user's userId, and if that id exists in the map, I combine the user's fields with the matching activity fields using array_merge() and push the merged row into \$joined. If there isn't a match, I just add the user row as-is.



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Section #4: (2.5 pts.) Misc

Progress: 100%

Task #1 (0.83 pts.) - Github Details

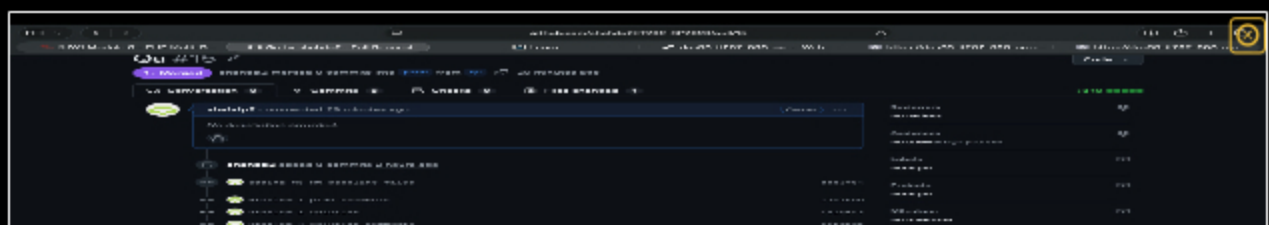
Progress: 100%

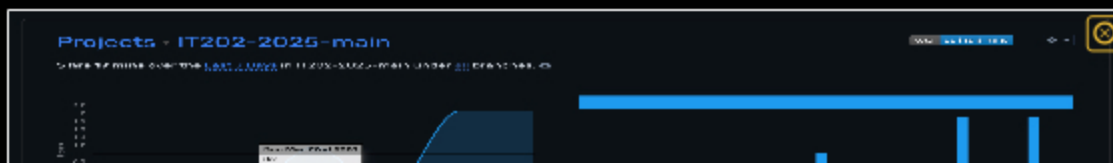
Part 1:

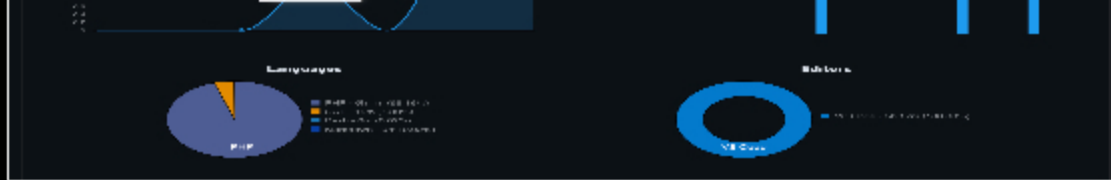
Progress: 100%

Details:

From the Commits tab of the Pull Request screenshot the commit history Following minimum should be present: addition of baseline files, comments planning thoughts, solutions







wakatime



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≡ Task #3 (0.83 pts.) - Reflection

Progress: 100%

⇒ Task #1 (0.33 pts.) - What did you learn?

Progress: 100%

Details:

Briefly answer the question with at least a few reasonable sentences

Your Response:

I learned how to work with multi-dimensional arrays in PHP and how to loop through them to extract only the fields I need. I also learned how to add new computed properties (like age and isClassic) based on existing values and today's date. Finally, I practiced joining two arrays by a shared key (userId) by building an index and merging matching rows.



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⇒ Task #2 (0.33 pts.) - What was the easiest part of the assignment?

Progress: 100%

Details:

Briefly answer the question with at least a few reasonable sentences

Your Response:

The easiest part was Problem 1 because it was mostly just looping through each bird and copying a few properties into a new array. It was straightforward to read the existing keys and build the subset array. Once the pattern was set, it was quick to repeat for each record.



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⇒ Task #3 (0.33 pts.) - What was the hardest part of the assignment?

Progress: 100%

Details:

Briefly answer the question with at least a few reasonable sentences

Your Response:

The hardest part was Problem 3 because joining arrays can get confusing if you don't organize it well. I had to make sure the matching was done by userId and that I didn't accidentally mismatch activities with the wrong user. Making the join clean (without nested loops getting messy) took the most thought.



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